Notes

what makes me confident
of what I say?

SOURCES AND ABBREVIATIONS

Sources cited in these notes are listed in the bibliography.
Abbreviations used in these notes are:

ECP-1—The Collected Papers of Albert Einstein, Volume 1, cited in bibliography as ECP-1.
ECP-2—The Collected Papers of Albert Einstein, Volume 2, cited in bibliography as ECP-2.
INT—Interviews by the author, listed at beginning of bibliography.
MTW—Misner, Thorne, and Wheeler (1973).

PROLOGUE

Page

23 [Of all the conceptions . . . finding them.] This paragraph is adapted from Thorne (1974).

26 [From the orbital period . . . ("10 solar masses").] Newton's formula is $M_h = C_o^3/(2\pi G P_o^2)$, where M_h is the mass of the hole (or any other gravitating body), C_o and P_o are the circumference and period of any circular orbit around the hole, π is 3.14159 . . . , and G is Newton's gravitation constant, 1.327×10^{11} kilometers^3 per second^2 per solar mass. See note to page 61, below. Inserting into this formula the starship's orbital period P_o = 5 minutes 46 seconds, and its orbital circumference $C_o = 10^6$ kilometers, one obtains a mass M_h = 10 solar masses. (One solar mass is 1.989×10^{30} kilograms.)

28 [As for size,. . . Sun's mass.] The formula for the horizon circumference is $C_h = 4\pi G M_h/c^2 = 18.5$ kilometers × $(M_h/M_\odot)$, where M_h is the hole's mass, G is Newton's gravitation constant (see above), $c = 2.998 \times 10^5$ kilometers per second is the speed of